

ARTIFICIAL INTELLIGENCE FOUNDATIONS

Seeing & Speaking

An overview of **Natural Language Processing & Computer Vision**

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Agenda

We will explore two of the most impactful pillars of modern AI: how machines understand human language and how they interpret the visual world.



Natural Language Processing (NLP)

Understanding and generating text.



Computer Vision (CV)

Classification, Detection, Segmentation, Tracking.

Natural Language Processing

NLP is the branch of AI that gives machines the ability to read, understand, and derive meaning from human languages.

The Goal

To bridge the gap between human communication and computer understanding.

The Challenge

Language is ambiguous, contextual, and constantly evolving (slang, idioms).

The Breakthrough

Transformer models (like BERT, GPT) revolutionized how we handle context.

Core Tasks in NLP



Sentiment Analysis

Determining the emotional tone behind words. Used heavily in brand monitoring and customer service.



Machine Translation

Automatically converting text from one language to another while preserving meaning.



Text Summarization

Shortening long documents into concise summaries that retain key information.



Chatbots & QA

Conversational agents that can understand queries and provide relevant answers from data.

PART 2

Computer Vision

Computer Vision is the field enabling computers to "see", identify, and process images in the same way that human vision does.

 Input: Pixels

 Output: Information

CORE TASK 1

Image Classification

"What is in this image?"

- ✓ Assigns a single label to the entire image.
- ✓ Simplest form of CV.
- ✓ **Example:** Is this an image of a Cat or a Dog?



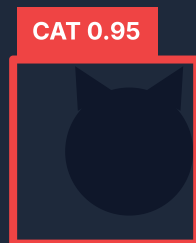
Label: Cat (98%)

CORE TASK 2

Object Detection

"Where are they?"

- ✓ Identifies objects AND their location.
- ✓ Uses **Bounding Boxes** (X, Y coordinates, Width, Height).
- ✓ **Example:** Autonomous vehicles detecting pedestrians and other cars.

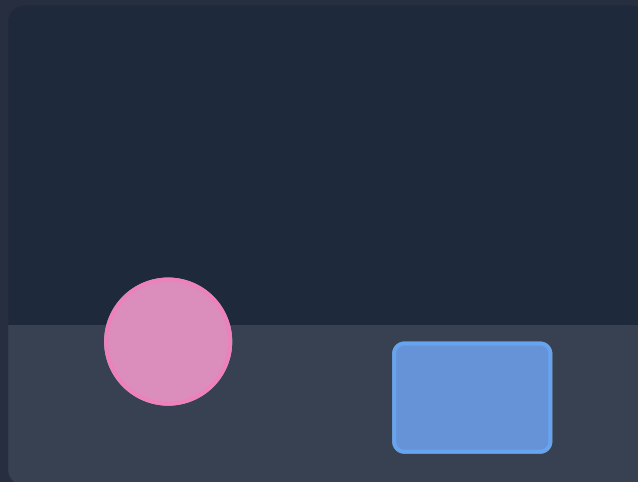


CORE TASK 3

Image Segmentation

"Which exact pixels belong to the object?"

- ✓ Pixel-level precision. Much more detailed than a bounding box.
- ✓ **Semantic Segmentation:** All "cars" are blue, all "roads" are gray.
- ✓ **Instance Segmentation:** Car #1 is blue, Car #2 is red (distinguishes individuals).
- ✓ **Example:** Medical imaging (identifying exact shape of a tumor).



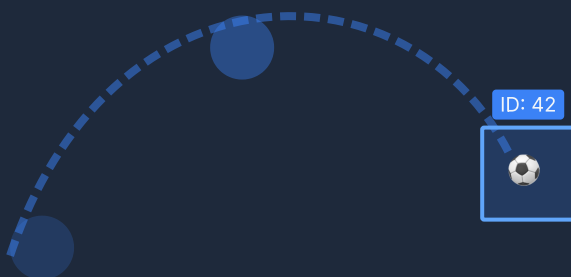
Masks applied to specific object pixels

CORE TASK 4

Object Tracking

"Where is it going?"

- ✓ Follows an object across multiple frames (video).
- ✓ Assigns a unique ID to keep track of specific instances even if they move or are briefly occluded.
- ✓ **Example:** Sports analytics (tracking a soccer ball), surveillance.



The Future is Multimodal

Modern AI increasingly combines NLP and Computer Vision.



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Vision

Language

Multimodal AI

- Image Captioning
- Visual Question Answering
- Generating images from text (e.g., DALL-E, Midjourney)

END OF PRESENTATION